week 3 2.02.2026

① Extensive-form games, SPE

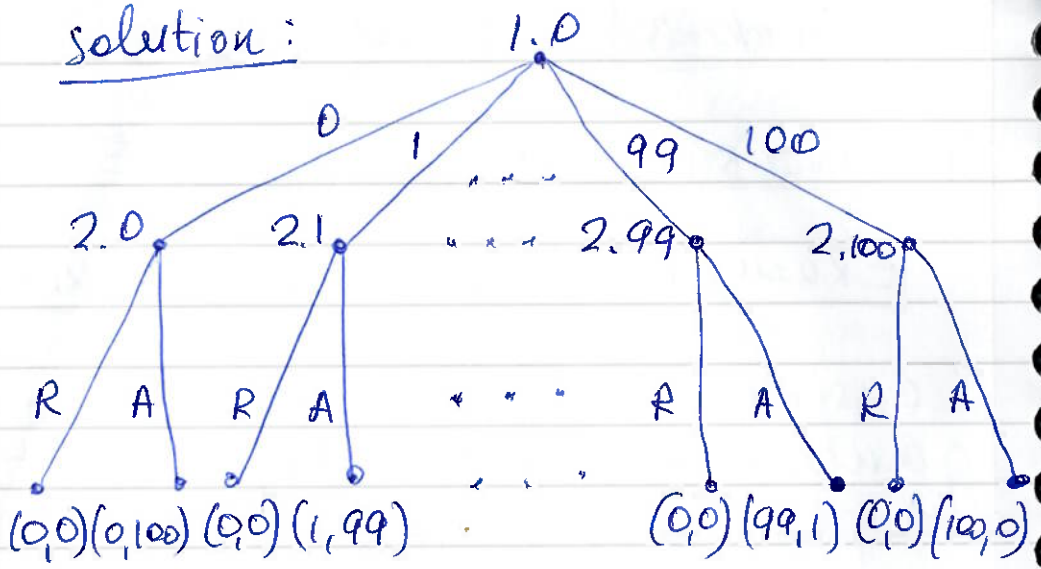
Example - ultimatum game

Consider the following two-stage game, about how 2 players split a pile of 100 gold coins.

The action (strategy) set of player 1 is given by $S_1 = \{0, 1, \dots, 100\}$, with choice i meaning that player 1 claims to keep i of the gold coins. Player 2 learns the choice of player 1, and then takes one of the two actions in response: A (accept) or R (reject). If player 2 plays accept, the payoff vector is $(i, 100-i)$. If player 2 plays reject, the payof vector is $(0,0)$.

(a) Describe the extensive-form version of the game using a tree representation.

solution:



(b) Describe the normal form of the game. It suffices to specify the strategy spaces and payoff functions.
(Hint: player 2 has 2^{101} pure strategies)

solution:

The strategy spaces are

$$S_1 = \{0, 1, \ldots, 99, 100\} \text{ and}$$

$$S_2 = \{0, 1\}^{101}, \text{ where}$$

the interpretation of

$$d = (d(i) : 0 \le i \le 100) \in S_2$$

is $d(i) = 1$ if player 2 accepts when player 1 plays i, and $d(i) = 0$ if player 2 rejects when player 1 plays i.

The payoff functions are given by $u_1(i, d) = i \cdot d(i)$

$$u_2(i, d) = (100 - i) \cdot d(i).$$

(c) Identify a NE of the normal-form game with payoff vector $(50, 50)$.

<u>Solution</u>: All profiles where $i = 50$ and $d(i) = 1$ give payoffs $(50, 50)$. However, for $i = 50$ to be a best response of player 1, we need player 2 to reject any claim $i > 50$.

Consider for example, the pair

of strategies $(50, d50)$ where

$d50(i) = 1$ for $i \leq 50$ and
$d50(i) = 0$ for $i > 50$.

That is, when player 2 uses the strategy $d50$, she accepts if player 1 plays $i \leq 50$ and she rejects otherwise. Obviously, the best choice for player 1 is to claim as much as they can, that is $i = 50$, and accepting is indeed a best response for player 2 as otherwise they will get 0. So, this is an NE. Is this an SPE?

(d) Identify the subgame-perfect equilibria of the extensive-form game. (Hint: There are 2 of them)

Solution: In the subgame beginning at node $2.i$, if $i \leq 99$, player 2 is strictly better off

to accept (payoff $100-i$) than to reject (payoff 0), so in a subgame perfect equilibrium, player 2 must accept whenever $i \leq 99$. The only degree of freedom left for player 2 is the value of $d(100)$, which could either be 0 or 1.

Let d_j be the strategy for player 2 such that she accepts if and only if i, the action of player 1, satisfies $i \leq j$. Then,

$(99, d_{99})$ and $(100, d_{100})$

are subgame-perfect equilibria.

② Repeated games

Example

Consider an infinitely repeated

game such that with probability p the game continues to the next round and with probability $(1-p)$ it ends.

The "trigger" strategy is such that a player cooperates as long as the other does, and defects forever after if the other player defects.

	C	D
C	4, 4	0, 5
D	5, 0	1, 1

1a) If the other player uses a trigger strategy, what is the total expected payoff from always cooperating?

<u>Solution:</u> If a player always cooperates, then given

that the other player uses grim trigger strategy, the other player always cooperates as well. Thus, each earns 4 every period, and the expected utility is

$$4+4p+4p^2+\dots$$

(b) What is the total expected payoff from always playing defect?

Solution: If player always defects, then given that the other player uses grim trigger strategy, the other player cooperates at round 1 but then defects in round 2 onwards. Thus, our player earns 5 in round 1 and 1 in every period from 2 onwards:

$$5+p+p^2+\dots$$

(c) Let p^* be the threshold such that when $p \geq p^*$, cooperation is sustainable as a subgame-perfect equilibrium by the grim trigger strategy. What is p^*?

<u>Solution</u>: The continuation payoff from indefinite cooperation is

$$4 + 4p + 4p^2 + \dots = 4 + \frac{4p}{1-p}$$

The continuation payoff from defecting (and thus continued defection) is

$$5 + p + p^2 + \dots = 5 + \frac{p}{1-p}$$

In order to sustain cooperation, we need

$$4 + \frac{4p}{1-p} \geq 5 + \frac{p}{1-p} \Rightarrow \boxed{p^* = \frac{1}{4}}$$